

2D Arrays & Pointers (Simple English Notes)

1. What is a 2D Array?

A 2D array looks like a table with rows and columns. But inside memory, the computer stores it as a single continuous list (row by row).

```
int a[3][3] = {
    {6, 2, 5},
    {0, 1, 3},
    {4, 9, 8}
};
```

2. Memory Layout

Even though it looks like a table, the actual memory is stored like this:

```
[6, 2, 5, 0, 1, 3, 4, 9, 8]
```

3. What does 'a' mean?

'a' points to the first row, not the first element. This is a key concept.

4. Understanding step by step

```
a      -> address of first row
*a     -> first row [6,2,5]
**a    -> first element 6
```

5. Pointer Access Examples

```
*(a + 1) -> 2
*(a + 1) + 2 -> 3
*(a + 2) + 0 -> 4
```

6. Important Formula

```
a[i][j] = (*(a + i) + j)
```

7. Simple Thinking Method

Step 1: Go to row Step 2: Go inside row Step 3: Pick value

8. Example

Find a[2][0]:

```
*(a + 2) + 0 = 4
```

9. Common Mistake

Many people think a is int**. This is wrong. It is actually a pointer to a row.

10. Using 1D pointer

```
int *p = &a[0][0];
```

```
p[0] = 6  
p[1] = 2  
p[2] = 5  
p[3] = 0
```

11. Practice Code

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int a[3][3] = {  
        {6,2,5},  
        {0,1,3},  
        {4,9,8}  
    };  
  
    cout << **a << endl;  
    cout << *(*a + 1) << endl;  
    cout << *(*a+1) + 2 << endl;  
    cout << *(*a+2) + 0 << endl;  
}
```